28 / 06 / 2022

EXPERIMENT - 07

3 BIT ASYNCHRONOUS UP/DOWN COUNTER

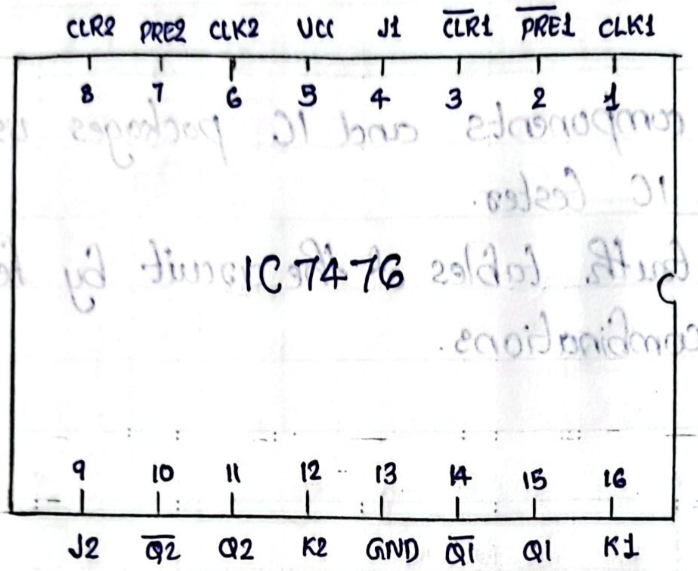
Aim: To design and verify 3-bit asynchronous up/down counter.

Components Required:

Sl-No	COMPONENT	SPECIFICATION	QTY
01	JK FLIP FLOP	IC 7476	2
02	XOR GATE	IC 7486	1
03	IC TRAINER KIT		1

Moved

Theory: A counter is a register capable of counting number of clock pulse arriving at its clock input. Counter represents the number of clock pulses arrived. A specified sequence of states appears as counter output. There are two types of counter, synchronous and asynchronous. In synchronous common clock is given to all flip flop and in asynchronous first flip flop is clocked by external pulse and then each successive flip flop is clocked by Q or Q output of previous stage. A soon the clock of second stage is triggered by output of first stage. Because of inherent propagation delay time all flip flops are not activated at same time which results in asynchronous operation.



M	Q_2	Q_1	Q_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

A three bit up/down counter from 0 to 7 when it is in upmode and from 7 to 0 when it is in down mode.

Procedure :-

(i) Connections are given as per circuit diagram
(ii) Logical inputs are given as per circuit diagram
(iii) Observe the output and verify the truth table.

Result :-

3-bit asynchronous counter was designed and truth table verified.

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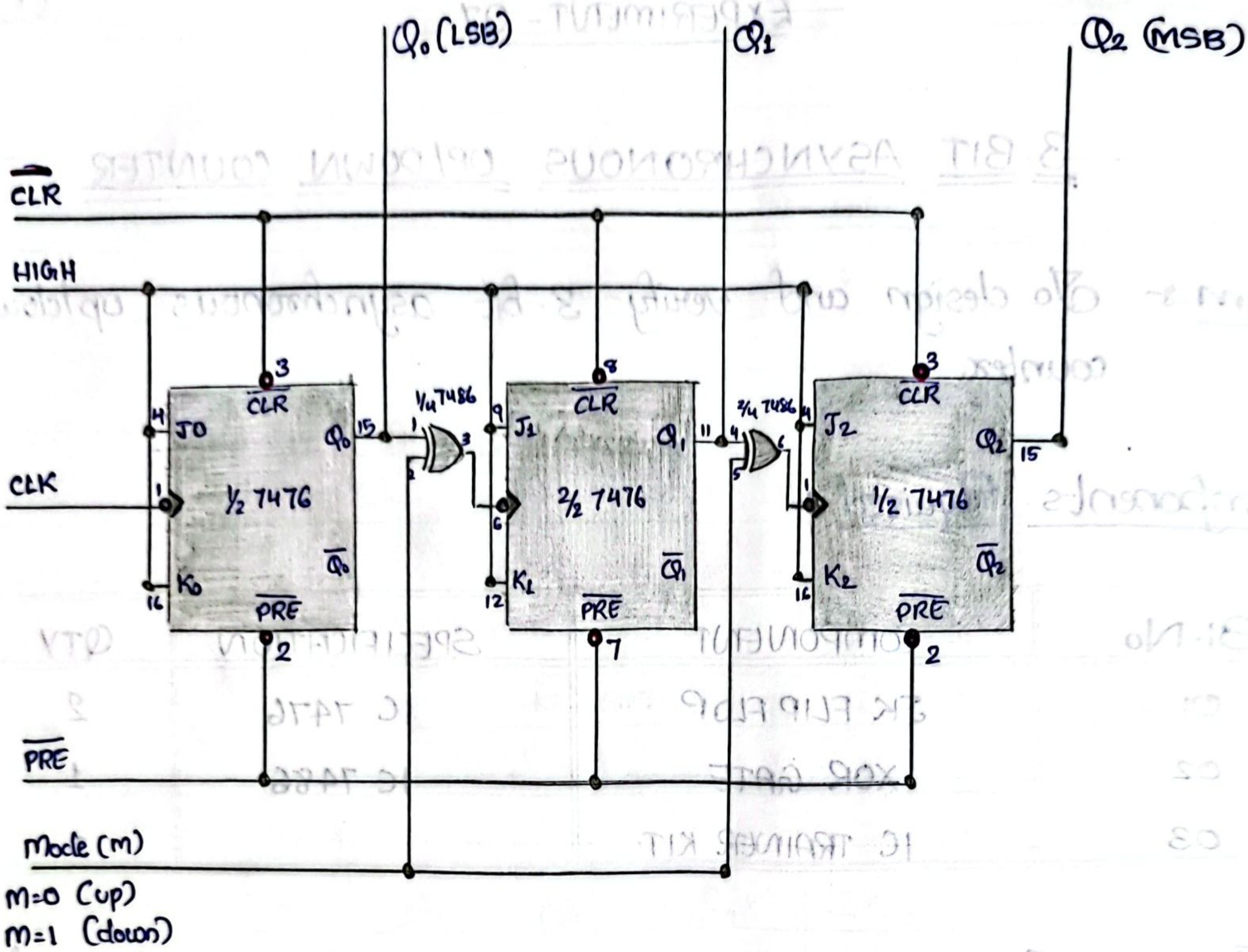

Q0 (LSB)
Q1
Q2 (MSB)
CLR
HIGH
CLK
PRE
Mode (m)
J0
K0
CLR
PRE
Q0
1/2 7476
1/4 7486
J1
K1
Q1
2/2 7476
2/4 7486
J2
K2
Q2
1/2 7476
M=0 (up)
M=1 (down)

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EXPERIMENT NO.8

SYNCHRONOUS COUNTER

Aim : To design and verify synchronous upcounter and mode-5 counter.

Components Required :

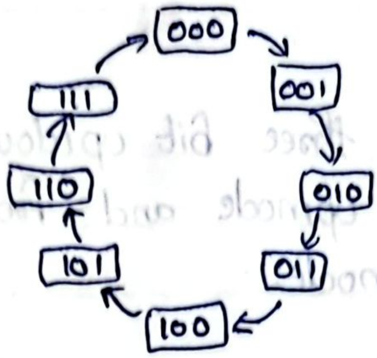
Sl.No	COMPONENT	SPECIFICATION	QTY
1	JK Flip Flop	IC 7476	2
2	2 I/P AND GATE	IC 7408	1
3	IC TRAINER KIT		1

Proceed

Theory :

A counter is a register capable of counting number of clock pulse arriving at its clock input. Counter represents the number of clock pulses arrived. An up counter is one that is capable of progressing in increasing order or decreasing order through a certain sequence. An up counter is operation is controlled by up signal. When this signal is high counter goes through up sequence. and when signal is low, counter follows reverse sequence.

SYNCHRONOUS UPCOUNTER



PRESENT STATE			NEXT STATE			A		B		C	
Q_A	Q_B	Q_C	Q_{A+1}	Q_{B+1}	Q_{C+1}	J_A	K_A	J_B	K_B	J_C	K_C
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

Q_A \ Q_BQ_C	00	01	11	10
0	0	0	1	0
1	X	X	X	X

$J_A = Q_BQ_C$

Q_A \ Q_BQ_C	00	01	11	10
0	X	X	X	X
1	0	0	1	0

$K_A = Q_BQ_C$

Q_A \ Q_BQ_C	00	01	11	10
0	0	1	X	X
1	0	1	X	X

$J_B = Q_C$

Q_A \ Q_BQ_C	00	01	11	10
0	X	X	1	0
1	X	X	1	0

$K_B = Q_C$

Q_A \ Q_BQ_C	00	01	11	10
0	1	X	X	1
1	1	X	X	1

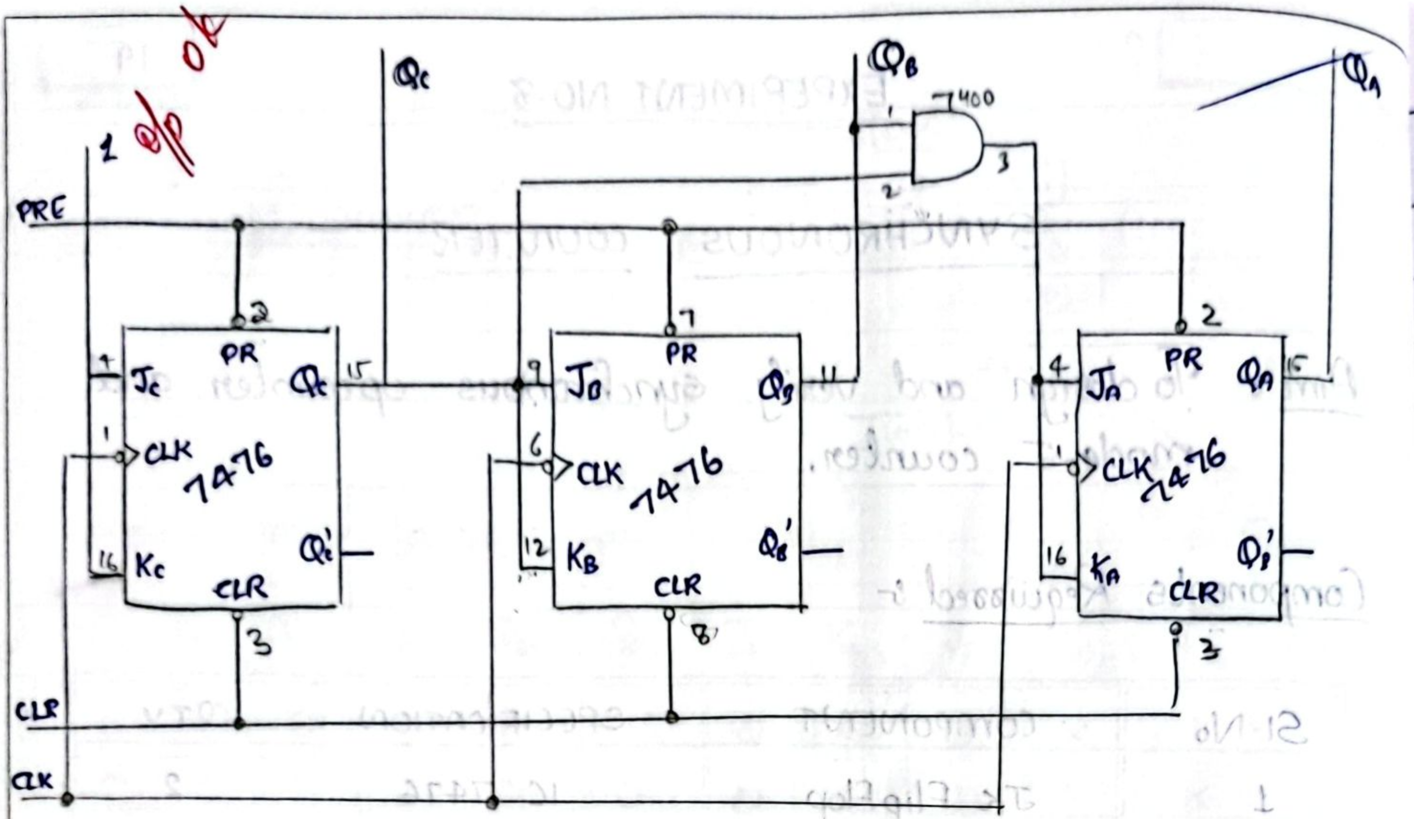
$J_C = 1$

Q_A \ Q_BQ_C	00	01	11	10
0	X	1	1	X
1	X	1	1	X

$K_C = 1$

Procedure

1) Connections are given as per circuit diagram.
2) Logical inputs are given as per circuit diagram.
3) Observe the output and verify the truth table.



MOD-5 COUNTER:

Truth Table

Q_C	Q_B	Q_A
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
0	0	0

Present count			Next count		
Q_C	Q_B	Q_A	Q_C	Q_B	Q_A
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	0	0	0

JK FF excitation table:

Q	Q+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Result :-

Synchronous counter and mod 5 counter is designed and verified with truth table.

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$Q_C \backslash Q_BQ_A$	00	01	11	10
0	1	X	X	1
1	0	X	X	X

$J_A = \overline{Q_C}$

$Q_C \backslash Q_BQ_A$	00	01	11	10
0	X	1	X	X
1	X	X	X	X

$K_A = 1$

$Q_C \backslash Q_BQ_A$	00	01	11	10
0	0	1	X	X
1	0	X	X	X

$J_B = Q_A$

$Q_C \backslash Q_BQ_A$	00	01	11	10
0	X	X	1	0
1	X	X	X	X

$K_B = Q_A$

$Q_C \backslash Q_BQ_A$	00	01	11	10
0	0	0	1	0
1	X	X	X	X

$J_C = Q_BQ_A$

$Q_C \backslash Q_BQ_A$	00	01	11	10
0	X	X	X	X
1	1	X	X	X

$K_C = 1$